

Non-Executing Security Analysis of AI Model Supply-Chain Artifacts

Provenance, Serialization, Impersonation, and Repository Risk Inspection

github.com/poojakira/nf-model-provenance-scanner - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

241

Tests passed

1

Tests skipped

6

Additional pytest subtests passed

75.81%

Statement coverage

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface

Data, models, requests, policies, or artifacts entering the tested path.

Control surface

Repository-specific validation, analysis, and fail-closed checks.

Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Enumerate repository and model artifacts.

02

Parse supported serialization/model formats.

03

Apply AST, pattern, symbolic, taint-style, metadata, and provenance checks.

04

Reject unsafe dynamic-execution paths in the scanner itself.

05

Emit findings and fail-loud states for incomplete or unsafe analysis.

Research Contribution

A primarily non-executing model-artifact security scanner combining static code analysis, symbolic and taint-style inspection, binary-format parsing, provenance checks, and fail-loud handling of incomplete analysis before model loading.

Verified Evidence

EV-01	241 passed, 1 skipped, 6 subtests passed.	EV-02	75.81% statement coverage; CI gate is 75%.
EV-03	Lint/format, type checking, Bandit, pip-audit, CodeQL, Trivy container scanning, unsafe-dynamic-execution rejection, and Docker build jobs succeeded.	EV-04	The committed adversarial fixture evidence remains fixture-scoped; it must not be generalized to arbitrary model repositories.
EV-05	CI coverage gate: 75%		

Method and Control Path

01	Enumerate repository and model artifacts.
02	Parse supported serialization/model formats.
03	Apply AST, pattern, symbolic, taint-style, metadata, and provenance checks.
04	Reject unsafe dynamic-execution paths in the scanner itself.
05	Emit findings and fail-loud states for incomplete or unsafe analysis.

Limitations and Residual Risk

- Static analysis cannot prove a model is safe.
- No claim is made that neural weight backdoors are generally detected.
- Fixture-level results do not establish population-level precision or recall.
- Unsupported or incomplete parsing must remain fail-loud rather than silently clean.

Reproducibility

Expected current-main evidence: 241 passed, 1 skipped, 6 subtests passed, 75.81% coverage.

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.
Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.
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PDF and PNG are generated locally from committed evidence without paid services.